

Q7 a) Write a program using function overloading to calculate the area of a laboratory (square/rectangle).

```
#include <iostream>
using namespace std;
float area(float r){ return 3.14f*r*r; }
int area(int s){ return s*s; }
int area(int l,int b){ return l*b; }
int main(){
    float r; int s,l,b;
    cin>>r>>s>>l>>b;
    cout<<area(r)<<"\n"<<area(s)<<"\n"<<area(l,b)<<"\n";
    return 0;
}
```

Q7 b) Write a program using function overloading to calculate the sum of float values and sum of integers.

```
#include <iostream>
using namespace std;
int add(int a,int b){ return a+b; }
float add(float a,float b){ return a+b; }
int main(){ int a,b; float x,y; cin>>a>>b>>x>>y; cout<<add(a,b)<<"\n"<<add(x,y)<<"\n"; return 0; }
```

Q7 c) Write a program to demonstrate Unary operator overloading when used with object so that when

```
#include <iostream>
using namespace std;
class Test{
public:
    void show(int x){ cout<<x<<"\n"; }
    void show(double y){ cout<<y<<"\n"; }
};
int main(){ Test t; t.show(10); t.show(10.55); return 0; }
```

Q7 d) Write a program to overload the '++' operator for one object of class counter.

```
#include <iostream>
using namespace std;
class Counter{
    int c;
public:
    Counter():c(0){}
    void operator++(){ ++c; }
    void display(){ cout<<c<<"\n"; }
};
int main(){
    Counter obj;
    ++obj;
    obj.display();
    return 0;
}
```

Q8 a) Write a program to overload '+' operator so that two strings can be concatenated.

```
#include <iostream>
#include <string>
using namespace std;
class Conc{
public:
    string s;
    void accept(){ getline(cin,s); }
```

```

    Conc operator+(Conc c2){
        Conc temp; temp.s=s+c2.s; return temp;
    }
    void display(){ cout<<s<<"\n"; }
};
int main(){
    Conc a,b,c;
    a.accept(); b.accept();
    c=a+b;
    c.display();
    return 0;
}

```

Q8 b) Write a program to create base class Login and derived class Employee to demonstrate runtime polymorphism.

```

#include <iostream>
#include <string>
using namespace std;
class Login{
public:
    string id,pass;
    void accept(){ getline(cin,id); getline(cin,pass); }
    virtual void display(){ cout<<id<<"\n"<<pass<<"\n"; }
};
class Employee: public Login{
public:
    string email;
    void acceptE(){ getline(cin,email); }
    void display(){ cout<<id<<"\n"<<pass<<"\n"<<email<<"\n"; }
};
int main(){
    Employee e;
    e.accept();
    e.acceptE();
    e.display();
    return 0;
}

```

Q9 a) Write a program to copy the contents of one file into another.

```

#include <fstream>
#include <string>
using namespace std;
int main(){
    ifstream in("First.txt");
    ofstream out("Second.txt");
    string line;
    while(getline(in,line)) out<<line<<"\n";
    in.close(); out.close();
    return 0;
}

```

Q9 b) Write a C++ program to count digits and spaces using file handling.

```

#include <fstream>
#include <string>
#include <cctype>
using namespace std;
int main(){

```

```

    ifstream in("input.txt");
    ofstream out("count.txt");
    string line; int count=0;
    while(getline(in, line)){
        for(char c:line) if(isdigit(c)) count++;
    }
    out<<count;
    return 0;
}

```

Q9 c) Write a C++ program to count words using file handling.

```

#include <fstream>
#include <string>
using namespace std;
int main(){
    ifstream in("input.txt");
    ofstream out("count.txt");
    string word; int count=0;
    while(in>>word) count++;
    out<<count;
    return 0;
}

```

Q10 a) Write a C++ Program to find Sum of Array elements using function template.

```

#include <iostream>
using namespace std;
template <class T>
T sum(T a[],int n){
    T s=0;
    for(int i=0;i<n;i++) s+=a[i];
    return s;
}
int main(){
    int a[5]={1,2,3,4,5};
    float b[3]={1.1,2.2,3.3};
    cout<<sum(a,5)<<"\n"<<sum(b,3)<<"\n";
    return 0;
}

```

Q10 b) Write a program to implement simple calculator using class template.

```

#include <iostream>
#include <string>
using namespace std;
template <class T>
class Demo{
public:
    T val1,val2;
    Demo(T a,T b){ val1=a; val2=b; }
    T sum(){ return val1+val2; }
};
int main(){
    Demo<int> d1(5,10);
    Demo<float> d2(3.2,2.1);
    cout<<d1.sum()<<"\n"<<d2.sum()<<"\n";
    return 0;
}

```

Q11 a) To read and display the value of a given vector.

```
#include <iostream>
#include <vector>
using namespace std;
int main(){
    vector<int> v={10,20,30};
    int sum=0;
    for(int x:v) sum+=x;
    cout<<sum<<"\n";
    return 0;
}
```

Q11 b) To multiply by a scalar value.

```
#include <iostream>
#include <vector>
using namespace std;
int main(){
    vector<int> v={10,20,30};
    int k; cin>>k;
    for(int &x:v) x*=k;
    for(int x:v) cout<<x<<" ";
    return 0;
}
```

Q11 c) To create a vector of ten elements from 10,20...100.

```
#include <iostream>
#include <vector>
using namespace std;
int main(){
    vector<int> v(10);
    for(int i=0;i<10;i++) v[i]=10*(i+1);
    for(int x:v) cout<<x<<" ";
    return 0;
}
```

Q12 a) Write C++ program using STL to implement vector.

```
#include <iostream>
#include <vector>
using namespace std;
int main(){
    vector<int> v={1,2,3,4,5};
    for(int x:v) cout<<x<<" ";
    return 0;
}
```

Q12 b) Write C++ program using STL to implement queue.

```
#include <iostream>
#include <queue>
using namespace std;
int main(){
    queue<int> q;
    for(int i=1;i<=5;i++) q.push(i);
    while(!q.empty()){
        cout<<q.front()<<" ";
        q.pop();
    }
}
```

```
    }  
    return 0;  
}
```